DATA ANALYTICS PROJECT REPORT

Product and Payment Analysis

Evaluating product revenue, quantity sold and payment-channel performance

Project Details	Information
Project	Product and Payment Analysis
Dataset	supermarket_sales_cleaned.csv
Tools Used	Python, Pandas, NumPy, Matplotlib, Seaborn and Jupyter Notebook
Report Type	Academic and Business Analysis Report

Prepared for academic and professional portfolio use

EXECUTIVE SUMMARY

The project analyses supermarket product and payment performance using transaction records. The supplied findings identify Health & Beauty as the highest-revenue product line and E-wallet as the leading payment method. Product demand is reported to vary significantly. The analysis combines revenue, quantity, payment and rating measures to help management understand which product categories contribute to sales and how customers complete purchases. Exact revenue totals, quantities and monthly values must be taken from the executed notebook.

1. INTRODUCTION

Retail performance depends on product demand, selling price, customer experience and payment convenience. Revenue alone does not explain the full picture: quantity sold indicates volume, ratings provide a customer-feedback perspective and monthly analysis reveals changes over time. This project uses these measures to develop a balanced view of supermarket product and payment performance.

2. PROJECT OBJECTIVES

- Identify product categories contributing the greatest revenue.
- Compare quantities sold across products or product categories.
- Measure revenue generated through different payment methods.
- Examine product-related customer ratings.
- Study monthly product sales patterns.
- Recommend inventory, promotion and payment strategies based on the results.

3. DATASET DESCRIPTION

The project uses the cleaned supermarket transaction dataset, specified as 2,000 records and 16 features. The analysis requires product/category fields, revenue or sales amount, quantity, payment method, rating and date/month information where available. Field names should be matched to the actual CSV headers before running calculations.

Attribute	Description
Dataset file	supermarket_sales_cleaned.csv
Records	2,000
Features	16
Analysis environment	Jupyter Notebook with Python libraries

4. DATA PREPARATION AND VALIDATION

Load the cleaned CSV with Pandas and inspect its columns, data types and missing-value profile. Confirm that revenue, quantity and rating columns are numeric. Validate that quantities and revenue values are within plausible ranges and inspect negative or zero values to determine whether they represent returns, corrections or invalid records. Standardize product and payment category labels. Parse transaction dates using a consistent date format and derive month/year fields only when the date field supports this. Check that monthly aggregation uses chronological ordering rather than alphabetical month labels. Retain return transactions appropriately and document whether the analysis reports gross sales, net sales or sales excluding returns.

5. EXPLORATORY AND STATISTICAL ANALYSIS

For product performance, calculate total revenue and total quantity for each product or product line. Add average selling price only where the relevant price field is reliable. For payment analysis, calculate transaction count, revenue and percentage contribution by payment method. For product ratings, calculate mean, median, standard deviation and rating count. For monthly analysis, group transactions by year and month, then compare revenue and quantity. If the dataset includes returns, report the chosen treatment explicitly and consider a separate return analysis.

6. VISUALIZATION AND INTERPRETATION

Visualization	Purpose and interpretation
Product Revenue	Rank product lines by total revenue. Clearly state whether revenue includes returns and use consistent currency units.
Quantity Sold	Compare total quantity sold by product or category. Quantity measures demand volume and should not be treated as equivalent to revenue.
Revenue by Payment Method	Aggregate revenue by payment method and compare both total revenue and transaction share. E-wallet is identified as the leading payment method in the supplied findings.
Product Ratings	Compare average rating and rating distribution by product line. Include the number of ratings in each group because small samples can distort averages.
Monthly Product Sales Trends	Plot monthly sales in chronological order. Use a consistent definition of sales and indicate whether the chart displays revenue, quantity or both.

7. KEY FINDINGS

- Health & Beauty is reported as the highest-revenue product line.
- E-wallet is reported as the dominant payment method.
- Product demand is reported to vary significantly across product categories.

8. BUSINESS INSIGHTS

Health & Beauty's reported revenue leadership indicates that this category is an important contributor to supermarket sales. The finding should be checked against both revenue and quantity: high revenue may result from high volume, higher unit prices or a combination. E-wallet dominance suggests that digital payment convenience is important to the customer base. Variation in product demand indicates that inventory and promotion decisions should be category-specific rather than uniform.

9. BUSINESS RECOMMENDATIONS

- Prioritize stock availability for high-revenue categories while monitoring stock turnover and expiry risk.
- Use quantity and revenue together to distinguish high-volume products from high-value products.
- Review low-performing categories for pricing, assortment, visibility or seasonal demand issues.
- Maintain secure and reliable E-wallet infrastructure while monitoring transaction failures.
- Use product ratings alongside sales to identify products that sell well but receive weaker feedback.
- Compare monthly trends across equivalent periods before making seasonal purchasing decisions.

10. LIMITATIONS AND FURTHER WORK

Exact product-level revenue, quantity, payment-method revenue, rating averages and monthly trends were not included in the supplied project summary. The report therefore does not invent numerical rankings beyond the stated Health & Beauty finding. Revenue comparisons can be affected by returns, discounts and inconsistent definitions of net sales. Monthly trends should account for incomplete months and changes in the dataset's date coverage.

11. CONCLUSION

Product and payment analysis links what customers buy with how they pay. The supplied results point to Health & Beauty as the strongest revenue category and E-wallet as the leading payment method. Combining verified revenue, quantity, rating and monthly measures will support better inventory planning, product selection and payment operations.

APPENDIX: REPRODUCIBILITY CHECKLIST

- Retain the original dataset and the cleaned dataset separately.
- Record the exact cleaning steps and the number of rows removed or modified.
- Run all notebook cells from top to bottom before exporting the final notebook.
- Label every chart with a clear title, axis names and units.
- Add exact calculated values to the report only after verifying them from the notebook output.

• Keep the analysis code, report and README together in the project repository.